Sound Active Attention Framework for Remote Sensing Image Captioning

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***Abstract*— Attention mechanism-based image captioning methods have achieved good results in the remote sensing field, but are driven by tagged sentences, which is called passive attention. However, different observers may give different levels of attention to the same image. The attention of observers during testing, then, may not be consistent with the attention during training. As a direct and natural human–machine interaction, speech is much faster than typing sentences. Sound can represent the attention of different observers. This is called active attention. Active attention can be more targeted to describe the image; for example, in disaster assessments, the situation can be obtained quickly and the corresponding disaster areas can be located related to the specific disaster. A novel sound active attention framework is proposed for more specific caption generation according to the interest of the observer. First, sound is modeled by mel-frequency cepstral coefficients (MFCCs) and the image is encoded by convolutional neural networks (CNNs). Then, to handle the continuity characteristic of sound, a sound module and an attention module are designed based on the gated recurrent units (GRUs). Finally, the sound-guided image feature processed by the attention module is imported into the output module to generate descriptive sentence. Experiments based on both fake and real sound data sets show that the proposed method can generate sentences that can capture the focus of human.**

***Index Terms*— Active attention, remote sensing image captioning, semantic understanding.**

I. Introduction

THE goal of high-resolution remote sensing image (HRRSI) captioning [1]–[4] is to extract the content features of an HRRSI and then generate a descriptive sentence. Different from other tasks in remote sensing image field [5]–[11], such as remote sensing image classification [12]–[15] and object detection [16]–[19], HRRSI captioning pays more attention to the high-level semantic understanding of remote sensing images from the perspective of human beings. Such a high-level perspective allows HRRSI captioning to serve for a variety of remote sensing studies and real applications, including image classification and image retrieval.

HRRSI captioning's great potential has inspired numerous researchers to explore the possibility of using a sentence to describe the content of HRRSI. According to whether the attention mechanism is used, image captioning methods can be divided into two groups: general image captioning methods [1], [2], [4] and attention-based image captioning methods [3]. General image captioning methods only utilize the feature of HRRSI to generate the descriptive sentence [1], [3], [4]. The main framework of these methods is based on a convolutional neural network (CNN) and a recurrent neural network (RNN). The CNN is used as a feature extractor and the RNN is used as a language generator which generates a sentence based on the feature of HRRSI. The CNN in the HRRSI captioning task is pretrained traditional networks that follow a pooling layer behind the convolution layer. The feature of the image in general image captioning methods is not changed in the process of sentence generation. In attention-based image captioning methods, different image feature inputs are obtained by constantly modifying the weights of image features in different locations during the process of generating each word.

The attention-based image captioning methods are concerned by more and more researchers [3], [20]–[22]. But, the aforementioned attention mechanism is learned by tagging sentences. The tagging sentences' semantic meaning may not be consistent with the goal of the observer. In the remote sensing field, an image is observed with a certain purpose of the observer in most cases. It would be a natural idea to take the initiative to pay attention to the area of interest. In the human brain, active attention generated by top-down signals is an important part of the brain's control mechanism [23]. From this phenomenon, an active attention mechanism is considered in this article. The introduction of this attention mechanism can make sentence generation more consistent with the observer's own intentions.

As shown in Fig. 1, there are many different objects in a remote sensing image (trees, playground, buildings, river, bridge, farmland, planes, terminal, and so on). It is difficult

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Fig. 1. Example of remote sensing images from RSICD data set. As shown in the figure, the objects and their relations are complex. There are many different objects in a remote sensing image (trees, playground, buildings, river, bridge, farmland, planes, terminal, and so on), which means it is difficult to determine the interested object of an observer.

to judge what is the interested object of an observer. Based on this fact, an intuitive idea is to add active attention guiding information for HRRSI captioning. In the process of implementation, *how to decide the format of active attention information is the first problem*. The active attention information in the brain is the brain waves. As the interaction between image and text, the most direct way is to use text information as the format of active attention information. But from a human point of view, if a person is asked to write a text as input, a certain text writing knowledge is needed to be learned by the person. It is not convenient to take text as input. On the other hand, sound as input has made great progress [24]. The expression of information in the form of waves is the common characteristic of sound and brain waves. Here, human sound is used directly as the guiding information for active attention.

Sound, a natural way for human beings to communicate with each other, has led to many related studies within computer vision field [24]–[30], such as image classification [31] and action recognition [32]. However, the sound studies mainly focus on video-related works [24], [27]–[31]. There are a few works based on the still image [29], [33] and remote sensing images [34]. Different from the sound of a whole sentence utilized in [34], the sound carried out herein is a single-word level. Image when facing a remote sensing image contains many ground objects, the observer can get the caption about boats by just saying "boats." If you want to know the caption about airport, the caption of related image part is shown to you by just saying "airport." Even though the sound can serve as active attention information in many tasks, an issue is that sound information about HRRSI captioning task is not available on-the-shelf. This motivates us to build a sound data set for active attention HRRSI captioning.

For the first time, a pattern of active attention is proposed to guide the caption generation for HRRSI. It is not a trivial thing to construct a sound data set for captioning. Asking volunteers to speak words is not only a time-consuming and laborious task but also requires a relatively quiet recording studio. So, the sound data set is constructed by sound generation application. Though these generated sounds are fake, they are meaningful for a research purpose. To further validate, a relatively small real sound data set is recorded by human beings. Our data set is publicly available in BaiduPan.[1]

[1]https://pan.baidu.com/s/1NHPJYtxi7Lr6bMILZDiD-A. Extraction code: 2m73

This article focuses on active attention HRRSI captioning. Instead of simply adding the acoustic wave of sound, such a problem is considered in a sound active attention view. The sound information is utilized as active attention information to "choose" the image information during the generation of the caption. Based on the guided sound data set, a novel sound active attention caption generation method is proposed. First, the sound is preprocessed to get the input for the following module of network. The mel-frequency cepstral coefficients' (MFCCs) [35] feature, which is a popular sound feature in speech processing, is extracted from the input sound. HRRSI is represented by the VGG16 pretrained on ImageNet. Then, the gated recurrent unit (GRU) is utilized to capture the feature of sound. Instead of using GRU directly, a framework is designed based on GRUs, which considers active attention of sound. Finally, an output GRU is followed to generate descriptive sentence for the input HRRSI.

Compared with the existing works, the main contributions of this article contain the following aspects.

1) Taking into account the guidance of sound, a novel sound active attention caption generation framework is proposed to capture the interest of an observer.
2) An attention GRUs module is proposed to capture the feature of sound and determine the final feature imported into the sentence generation module.
3) A sound-guided data set for HRRSI captioning is first built based on the interest of observers.

The remainder of this article is organized as follows. Section II discusses the related works of HRRSI captioning and the sound studies in computer vision field. Section III describes the proposed sound active attention caption generation framework. The experimental results on three HRRSI captioning data sets are shown in Section IV. Finally, Section V concludes this article.

II. Related Work

A. Remote Sensing Image Captioning

There are a few studies about HRRSI captioning in the past few years [1]–[4], which mostly tackle the problem by utilizing deep-learning techniques. Depending on whether the attention mechanism is used, captioning methods are divided into two groups: general image captioning methods and attention-based image captioning methods.

The general image captioning methods are reviewed as follows. Qu *et al.* [1] represented the HRRSI by the pretrained CNN, then the image feature is imported into an RNN or long short-term memory (LSTM) to generate the descriptive sentence. Lu *et al.* [3] studied the captioning performance of several handcrafted features, including scale-invariant feature transform (SIFT) [36], bag of words (BOW) [37], Fisher vector (FV) [38], and vector of locally aggregated descriptors (VLADs) [39]. Shi and Zou [2] proposed a method based on fully convolutional networks [40], which detects the objects in the image and then integrates the detection results into predefined sentence templates to generate language descriptions.

In addition to general image captioning methods, attention-based image captioning methods are becoming

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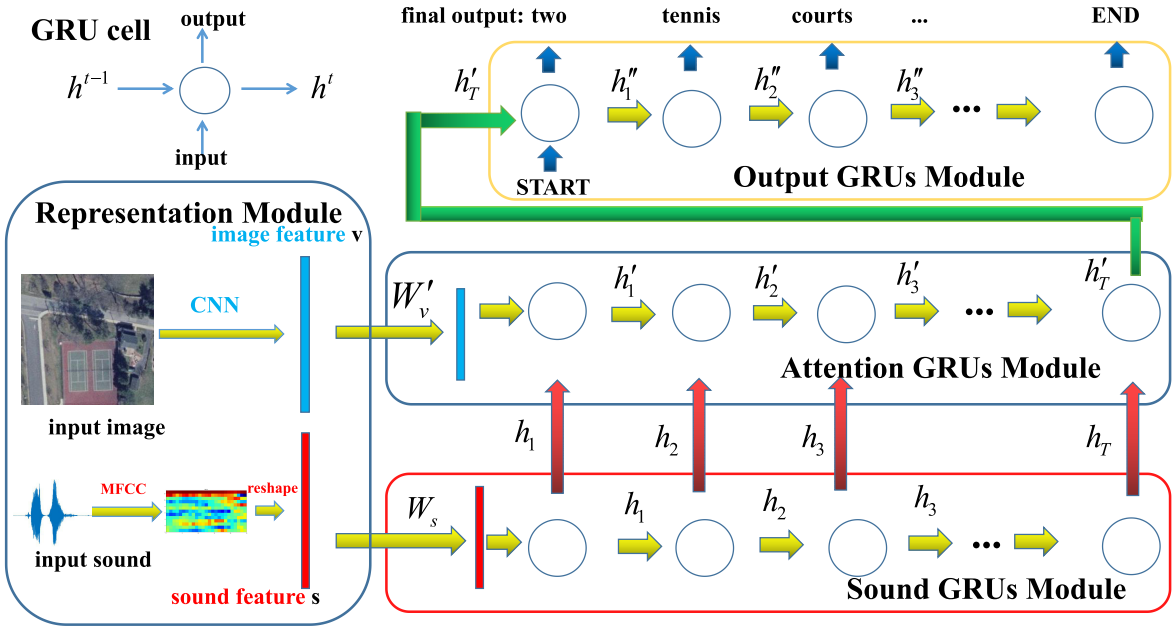


Fig. 2. Framework of the proposed method. There are four modules: representation module, sound GRUs module, attention GRUs module, and output GRUs module. First, the sound features of the representation module are encoded by the sound GRUs module. Second, the image features of the representation module and the hidden outputs of the sound GRUs module are encoded by the attention GRUs module. Finally, the output GRUs module generates a sentence by taking a sentence start token and the last steps' output of the attention GRUs module.

more and more popular. Lu *et al.* [3] introduced two kinds of attention mechanism ("hard" attention and "soft" attention) into remote sensing image captioning field and obtained amazing results. The attention parameters in "hard" attention and "soft" attention are computed by the hidden states of LSTM. Aneja *et al.* [41] proposed another attention mechanism in which attention parameters are computed by using convolutional layer activations. Wang *et al.* [42] proposed a captioning method which can provide more accurate object location by adding additional location and category tags to images. Zhang *et al.* [43] treated the information from the fully connected layer of the CNN as attribute information, which is used to improve the performance of captioning. Different from pervious works which try to generate a sentence given an image, work in this article is to generate a descriptive sentence considering the image and the specific sound at the same time. The generated sentence of the proposed method can capture the topics that an observer is interested in.

B. Sound Studies in Computer Vision Field

Sound is a natural way for human to express ideas and receive information. As a heated research aspect in the field of artificial intelligence, sound processing in computer vision field is getting more and more attention in recent years [24], [27]–[31], [33], [44]–[46]. When hitting on something, the physical properties of the material give impact on the generated sound. Considering this characteristic, Owens *et al.* [27] proposed an algorithm to generate a waveform from the input of silent videos. Instead of generating sound, Arandjelovic and Zisserman [29] aimed to localize the sound in an image. Different from sound source localization, Gao *et al.* [30] proposed an unsupervised training pipeline to separate a mixed sound from different sounds according to the objects. Apart from above works, Owens and Efros [24] tried to learn a fused multisensory representation by considering both audio and visual information. The representation is evaluated on several related tasks, such as action recognition and sound source localization. The introduction of sound gives a better representation for the action recognition performance [24]. A similar work is proposed by Chung *et al.* [33], and the task in [33] is to generate a video based on the inputs, which are a still face image and a speech segment. The task in this article is to generate a sentence based on the still remote sensing image and a segment sound for a specific word.

III. Sound Active Attention Framework for Remote Sensing Image Captioning

The sound active attention framework for the remote sensing image captioning method is proposed in this section, which mainly consists of four modules. As shown in Fig. 2, in the representation module, the image feature of input image is extracted by the CNN and the sound is represented by the feature of MFCC. Then, the sound feature is imported into the sound GRUs module. The output of sound GRUs module's every step is used to guide the attention GRUs module. Finally, the output of attention GRUs module is imported into output GRUs module to generate the final descriptive sentence.

In the traditional captioning methods, a sentence is generated by maximizing the probability of sentence y given the

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image I

$$\arg\max_{\psi} \sum_{(y,I)} \log p(y|I;\psi) \tag{1}$$

where y is the sentence and ψ denotes the parameters of the model. In this article, the task is changed to

$$\arg\max_{\psi} \sum_{(y,I,.\text{wav})} \log p(y|I,.\text{wav};\psi) \tag{2}$$

where y is the sentence, .wav denotes the sound file, and ψ denotes the parameters of model.

A. Representation Module

The representation module mainly comprised of three parts: image representation, sound representation, and sentence representation. The visual representation of remote sensing image I is extracted by CNNs, which is VGG-16 in this article. The output of the fully connected layer before the softmax is used to represent the visual representation

$$\mathbf{v} = \text{VGG16}(I) \tag{3}$$

where $\mathbf{v}$ denotes the visual representation.

As for sound file .wav, the MFCC is a popular feature in traditional speech signal processing [35]. The output of MFCC is reshaped into a vector as the input of the sound GRUs module

$$\mathbf{s} = \text{MFCC}(.\text{wav}) \tag{4}$$

where $\mathbf{s}$ denotes the sound representation.

For a descriptive sentence, the word is represented by a one-hot vector, whose dimension is the size of dictionary D. The one-hot vector is embedded into a vector with a fixed dimension. The length of the sentence is fixed to T. The sentence longer than T is filtered out during experiments and sentence shorter than T is padded by zero.

B. Sound GRUs Module

Sound is a guided information needed to model. A sound GRUs module is designed to capture the sound feature due to the GRU's good performance in sound modeling. GRU [47] is a gating mechanism which is used for machine translation first. The performance of GRU is proved experimentally better than LSTM to deal with the modeling of speech signal on smaller data sets [48]. Thus, a GRU layer is used to model the input sound information. The main structure of the GRU consists of two gates: update gate z_t and reset gate r_t. At step t, the previous output is h_{t-1}, and the output of step t is h_t. Initially, for $t = 0$, $h_0 = 0$. The input sound $\mathbf{s}$ is just imported to this module in the first step. The specific equations are as follows:

$$z_1 = \sigma(W_z W_s \mathbf{s} + b_z) \tag{5}$$

$$\tilde{h}_1 = \tanh(W_h \mathbf{s} + b_h) \tag{6}$$

$$h_1 = (1 - z_1) \circ \tilde{h}_1 \tag{7}$$

where W_s denotes a fully connected layer which changes the dimension of sound representation to the dimension of hidden state (128 in this article). W_z, W_h, b_z, and b_h are the learned parameters in the sound GRUs module. $\circ$ denotes the Hadamard product and σ is the hard sigmoid function

$$\sigma(x) = \max\left(0, \min\left(1, \frac{x+1}{2}\right)\right). \tag{8}$$

In the following recurrent steps, the hidden state h_t ($t > 1$) only depends on the previous hidden state h_{t-1}:

$$z_t = \sigma(U_z h_{t-1} + b_z) \tag{9}$$

$$r_t = \sigma(U_r h_{t-1} + b_r) \tag{10}$$

$$\tilde{h}_t = \tanh(U_h(r_t \circ h_{t-1}) + b_h) \tag{11}$$

$$h_t = z_t \circ h_{t-1} + (1 - z_t) \circ \tilde{h}_t \tag{12}$$

where U_z, U_r, b_r, and U_h are the learned parameters in the sound GRUs module. Other symbols have the same meaning as previously mentioned, which are omitted due to the repetition.

The output of every step in this module is imported to the attention GRUs module to get the more suitable feature for captioning task.

C. Attention GRUs Module

Attention GRUs module is a feature enhancement layer that can make the image feature trend to the sound-guided part. The inputs of this module include two parts: image visual representation $\mathbf{v}$ and the hidden state of the sound GRUs module h_t. The image visual representation $\mathbf{v}$ is first changed into the dimension of the hidden state of GRUs by a fully connected layer. For the attention GRUs module, when $t = 0$, $h'_0 = h_1$. The image visual representation is imported into this model only in the first step. The specific equations are as follows:

$$z'_1 = \sigma\left(W'_z\left(W'_v \mathbf{v} + h_1\right) + U'_z h'_0 + b'_z\right) \tag{13}$$

$$r'_1 = \sigma\left(W'_r h_1 + U'_r h'_0 + b'_r\right) \tag{14}$$

$$\tilde{h}'_1 = \tanh\left(W'_h\left(W'_v \mathbf{v} + h_1\right) + U'_h\left(r'_1 \circ h'_0\right) + b'_h\right) \tag{15}$$

$$h'_1 = z'_1 \circ h'_0 + \left(1 - z'_1\right) \circ \tilde{h}'_1 \tag{16}$$

where W'_v denotes a fully connected layer which changes the dimension of image visual representation to the dimension of hidden state (128 in this article). U'_z, U'_r, U'_h, W'_z, b'_z, W'_h, and b'_h are the learned parameters in the attention GRUs module.

In the following steps of this module, the hidden state h'_t ($t > 1$) depends on the previous hidden state h'_{t-1} and the corresponding hidden state h_t from the sound GRUs module

$$z'_t = \sigma\left(W'_z h_t + U'_z h'_{t-1} + b'_z\right) \tag{17}$$

$$r'_t = \sigma\left(W'_r h_t + U'_r h'_{t-1} + b'_r\right) \tag{18}$$

$$\tilde{h}'_t = \tanh\left(W'_h h_t + U'_h\left(r'_t \circ h'_{t-1}\right) + b'_h\right) \tag{19}$$

$$h'_t = z'_t \circ h'_{t-1} + \left(1 - z'_t\right) \circ \tilde{h}'_t \tag{20}$$

where U'_z, W'_r, U'_r, b'_r, and U'_h are the learned parameters in the attention GRUs module.

The image visual representation $\mathbf{v}$ is passed through all the steps of GRUs. The hidden state of the sound GRUs module h_t is utilized to compute the update gate and reset gate. The output of the gates is between 0 and 1, which is an embodiment of different attention weights. This gate mechanism provides

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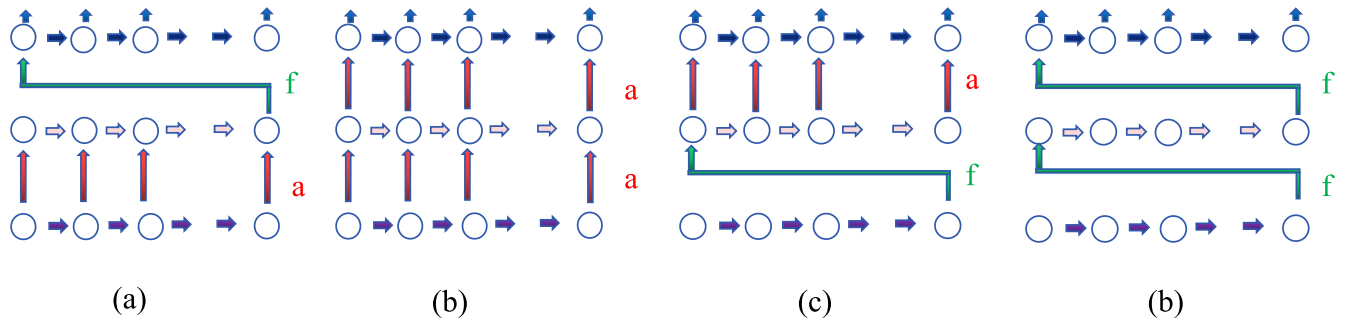


Fig. 3. Variants of the proposed framework. The "a" in "a-f," "a-a," and "f-a" means all the outputs of every step are imported into the next module and the "f" in "a-f," "f-a," and "f-f" means the first step's input is the output of the last step in the previous module. (a) a-f. (b) a-a. (c) f-a. (d) f-f.

an attention mechanism for image visual representation, which determines which part of the information can be transmitted via GRU. As the step goes on, the feature of sound and visual is combined together for the following sentence generation.

D. Output GRUs Module

The output GRUs module is designed to generate sentence. The input of this module is the output hidden state of the last step of the attention GRUs module. At the first step, a start token "START" ("START" is also included in dictionary) is embedded into a 128-dimension vector, which is the h_0'' in this module. Then, the hidden state h_1'' is mapped to dictionary size and a word w_1 is generated. The output of attention GRUs module's last step is denoted as h_T'. T is the step number the GRU repeated, which is also the number of word in a sentence. The specific equations are as follows:

$$z_1'' = \sigma\left(W_z'' h_T' + U_z'' h_0'' + b_z''\right) \tag{21}$$
$$r_1'' = \sigma\left(W_r'' h_T' + U_r'' h_0'' + b_r''\right) \tag{22}$$
$$\tilde{h}_1'' = \tanh\left(W_h'' h_T' + U_h''\left(r_1'' \circ h_0''\right) + b_h''\right) \tag{23}$$
$$h_1'' = z_1'' \circ h_0'' + \left(1 - z_1''\right) \circ \tilde{h}_1'' \tag{24}$$
$$w_1 = \text{softmax}\left(W_o h_1''\right) \tag{25}$$

where W_z'', U_z'', b_z'', W_r'', U_r'', b_r'', W_h'', U_h'', and b_h'' are the learned parameters in the output GRUs module. W_o represents a fully connected layer which converts the dimension of h_1'' to the size of dictionary. w_1 represents the first word of the generated sentence.

After the first word is generated based on h_T', the following words are generated based on the previous hidden state until meet the end token "END":

$$z_t'' = \sigma\left(U_z'' h_{t-1}'' + b_z''\right) \tag{26}$$
$$r_t'' = \sigma\left(U_r'' h_{t-1}'' + b_r''\right) \tag{27}$$
$$\tilde{h}_t'' = \tanh\left(U_h''\left(r_t'' \circ h_{t-1}''\right) + b_h''\right) \tag{28}$$
$$h_t'' = z_t'' \circ h_{t-1}'' + \left(1 - z_t''\right) \circ \tilde{h}_t'' \tag{29}$$
$$w_t = \text{softmax}\left(W_o h_t''\right) \tag{30}$$

where U_z'', b_z'', U_r'', b_r'', U_h'', and b_h'' are the learned parameters in the output GRUs module. W_o represents a fully connected layer which converts the dimension of h_t'' to the size of dictionary. w_t represents the tth word of the generated sentence.

To train the proposed model, the loss function that needs to be minimized is defined as follows:

$$L = -\sum_{t=0}^{T-1} y_{t+1} \log(w_{t+1}) + \lambda \sum_{\psi} \|\psi\|^2 \tag{31}$$

where y_{t+1} is the word in the ground-truth sentence and w_{t+1} is the corresponding predicted word. The second part of the loss function is the regularization of the weights in the proposed model, which is utilized to prevent overfitting. ψ is the weights in the proposed model. λ is the coefficient of regularization.

As shown in Fig. 3, there are three variants in the proposed framework. The framework described in detail earlier is denoted as "a-f." The "a" means that all the hidden states of the previous module are imported into the next module's hidden states. The "f" means that the output of the last step of the previous module is only imported into the first step of this module. These two conditions make four variants between three modules.

IV. Experiments

A. Data Sets and Competitors

1) Data Sets: To demonstrate the effectiveness of the proposed method, three HRRSI captioning data sets are included in experiments: UCM-captions, Sydney-captions, and Remote Sensing Image Captioning Data Set (RSICD).

a) UCM-captions: This data set, based on the UC Merced Land Use Data Set [49], is proposed in [1]. The images in UC Merced Land Use Data Set were manually cropped into 256 × 256 from many large images (whose spatial resolution is 0.3048 m) from the United States Geological Survey (USGS) National Map Urban Area Imagery. About 21-class land use images, including agricultural, airplane, baseball diamond, beach, building, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tank, and tennis court, are collected into UCM-captions with 100 images for each class. Based on [1], each image is described by five different sentences.

b) Sydney-captions: This data set, based on the Sydney Data Set [50], is also provided in [1]. A large image of Sydney, Australia, whose pixel resolution is 0.5 m, was got from Google Earth. After cropping and selecting, seven classes

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(613 images in total), including residential, airport, meadow, rivers, ocean, industrial, and runway, are remained to construct Sydney-captions. Five different sentences were used to explore the content of each image [1].

c) RSICD: This data set is provided in [3]. To construct RSICD, more than 10 000 remote sensing images, fixed to 224×224, were collected from Google Earth, MapABC, Tianditu, and Baidu Map with various resolutions. About 10 921 images are used to construct the data set after selection and cutting. According to the instructions considering the characteristics of remote sensing image [3], five sentences are given by different persons to describe a remote sensing image.

d) Sound Data Set: There is currently no image caption data set with sound information. The manual sound recording requires a relatively quiet recording place, and the recording needs a lot of manpower and material resources. To verify the proposed thoughts, a lot of experiments are performed first on the fake data sets, and the fake data sets are built as follows. The guided sounds of the above data sets are generated by sound generation application. The origin texts of sound are picked up from ground-truth descriptive sentence after screening and processing. First, the punctuation is removed. Then, after filtering out the English stop words, the nouns are picked up as the semantic words. Finally, the semantic words are tuned to sound file in format .wav. To further validate the utility of the method, a real sound data set is manually recorded based on UCM-captions data set.

2) Competitors: To verify the performance of the proposed method, comparative experiments are conducted on five captioning methods. The main framework of these methods is based on encoder–decoder. The difference between these methods is the choice of encoder or decoder and the coding implement. The five comparative captioning methods are denoted as: VLAD+RNN [3], VLAD+LSTM [3], mRNN [1], mLSTM [1], mGRU [51], mGRU-embedword, convcap [41], soft-attention [3], and hard-attention [3]. The details of these captioning methods are as follows.

a) VLAD+RNN: The encoder of image in this method is VLAD, a classic handcraft feature. The decoder is a naive RNN. The implementation of this method is based on the origin Python without using any deep-learning framework.

b) VLAD+LSTM: The encoder of image in this method is VLAD, a classic handcraft feature. The decoder is LSTM, a variant of naive RNN. The implementation of this method is based on the origin Python without using any deep-learning framework.

c) mRNN: The encoder of image in this method is the CNN, specifically VGG-16. The decoder is a naive RNN. The implementation of this method is based on Keras, one of the most popular deep-learning frameworks.

d) mLSTM: The encoder of image in this method is the CNN, specifically, VGG-16. The decoder is LSTM, a variant of the naive RNN. The implementation of this method is based on Keras, one of the most popular deep-learning frameworks.

e) mGRU: The encoder of image in this method is the CNN, specifically, VGG-16. The decoder is GRU, a variant of the naive RNN. The implementation of this method is based on Keras, one of the most popular deep-learning frameworks.

f) mGRU-embedword: The architecture of mGRU-embedword is the same as that of mGRU. The difference is that the word embedding layer is not trained from scratch. Instead, the word is embedded into a vector using a pretrained word model called global vector (GloVe) [52].

g) convcap: The encoder of image in this method is the CNN, specifically, VGG-16. The attention parameters are computed by using convolutional layer activations. The decoder is CNNs designed in [41]. The implementation of this method is based on PyTorch, one of the most popular deep-learning frameworks.

h) Soft-attention: The encoder of image in this method is the CNN, specifically, VGG-16. The decoder is LSTM, a variant of the naive RNN. The feature maps of *conv5_3* sized $14 \times 14 \times 512$ are used. The attention parameters are computed by using hidden states of LSTM. Different parts of feature maps are given different weights to decide the attention part. The implementation of this method is based on Theano, one of the most popular deep-learning frameworks.

i) Hard-attention: The encoder of image in this method is the CNN, specifically, VGG-16. The decoder is LSTM, a variant of the naive RNN. The feature maps of *conv5_3* sized $14 \times 14 \times 512$ are used. The attention parameters are computed by using hidden states of LSTM. The sampling strategy is used to decide the attention part. The implementation of this method is based on Theano, one of the most popular deep-learning frameworks.

B. Metrics of Remote Sensing Image Captioning

The metrics adopted for remote sensing image captioning are BiLingual Evaluation Understudy (BLEU) [53], Recall-Oriented Understudy for Gisting Evaluation (ROUGE_L) [54], Metric for Evaluation of Translation with Explicit ORdering (METEOR) [55], Consensus-based Image Description Evaluation (CIDEr) [56], and semantic propositional image caption evaluation (SPICE) [57].

1) BLEU: For BLEU, the co-occurrences of n-gram between the generated sentence and the reference sentence are measured, where n-gram is a set of one or more ordered words. Modified n-gram precision p_n defined in [53] is utilized to measure the matching of sentences. A multiplicative brevity penalty factor is introduced to add penalize to the candidate sentences, whose lengths are shorter than reference sentences. The brevity penalty BP is defined as

$$\mathrm{BP} = \begin{cases} 1, & \text{if } c > r \\ e^{(1-r/c)}, & \text{if } c \leq r \end{cases} \tag{32}$$

where c is the length of the candidate sentence and r is the effective reference sentence length. Then

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{33}$$

where p_n is the modified n-gram precision, N in this article is one to four, w_n is the positive weights, and the sum of w_n is one.

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2) ROUGE_L: ROUGE stands for Recall-Oriented Understudy for Gisting Evaluation. Given the length of the longest common subsequence (LCS), ROUGE_L is a metric calculating F-measure between X of length m and Y of length n. X is a reference sentence of an image and Y is a candidate sentence generated by computer

$$R_{\text{lcs}} = \frac{\text{LCS}(X,Y)}{m} \tag{34}$$

$$P_{\text{lcs}} = \frac{\text{LCS}(X,Y)}{n} \tag{35}$$

$$\beta = \frac{P_{\text{lcs}}}{R_{\text{lcs}}} \tag{36}$$

$$\text{ROUGE_L} = \frac{(1+\beta^2)R_{\text{lcs}}P_{\text{lcs}}}{R_{\text{lcs}}+\beta^2 P_{\text{lcs}}} \tag{37}$$

where $\text{LCS}(X,Y)$ is the LCS of X and Y.

3) METEOR: METEOR stands for Metric for Evaluation of Translation with Explicit ORdering which is measured by generating an alignment between the reference sentence and the generated sentence. Then, the unigram precision (P) and the unigram recall (R) are computed. Harmonic mean of P and $9R$ is computed as Fmean

$$\text{Fmean} = \frac{10PR}{R+9P}. \tag{38}$$

As Fmean is computed based on unigram precision and recall, a penalty is designed to take the longer matches into account

$$\text{Penalty} = 0.5\left(\frac{\text{chunks}_{\text{num}}}{\text{unigrams_matched}_{\text{num}}}\right)^3 \tag{39}$$

where the numerator $\text{chunks}_{\text{num}}$ is the number of chunks and the denominator $\text{unigrams_matched}_{\text{num}}$ is the number of matched unigrams.

Finally, the METEOR is computed as follows:

$$\text{METEOR} = \text{Fmean}(1-\text{Penalty}). \tag{40}$$

4) CIDEr: CIDEr is calculated by adding a term frequency inverse document frequency (TF-IDF) weighting $g_k(\cdot)$ for every n-gram. CIDEr is based on consensus, which means the reference sentences are more than one. X_j is the jth reference annotation of an image and Y is a candidate sentence generated by computer. CIDEr_{n} for n-grams is computed as follows:

$$\text{CIDEr}_{\text{n}}(Y,X) = \frac{1}{m}\sum_{j=1}^{m}\frac{g_n(Y)\cdot g_n(X_j)}{\|g_n(Y)\|\|g_n(X_j)\|} \tag{41}$$

where m is the number of conference sentences.

The longer n-grams are used to capture richer semantics and grammatical properties. CIDEr is a weighting sum of all the CIDEr_n

$$\text{CIDEr}(Y,X) = \sum_{n=1}^{N} w_n \text{CIDEr}_n(Y,X) \tag{42}$$

where $w_n = 1/N$ and $N = 4$.

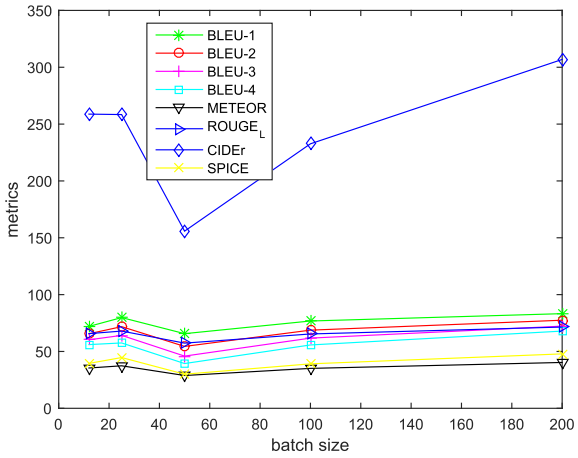


Fig. 4. Captioning metrics versus different sizes of training batch.

5) SPICE: SPICE stands for Semantic Propositional Image Caption Evaluation. Different from previous evaluation methods, which is based on the n-gram overlap of the generated sentences and the reference sentences, SPICE estimates generated sentence by transforming the sentences into a graph-based semantic representation. Experiments show that SPICE outperforms previous evaluations methods in terms of consistency with human evaluations.

For all these metrics, the larger they are, the better the generated sentence is.

C. Experimental Setup

In the experiment, the same splitting form is followed for three HRRSI captioning data sets: 80% for training, 10% for evaluation, and 10% for test. The implementation is based on the Keras deep-learning framework. The loss function is optimized by an efficient optimization algorithm called Adam [58]. The value of the regularization coefficient, λ, is set to 10^{-8}. During the calculation of the MFCC feature, the default parameters are adopted in package python_speech_features.[2] The longest sentence length T is set to 30. The sizes of dictionary D for different date sets are as follows: 1702 for RSICD, 308 for UCM-captions, and 199 for Sydney-captions. The model is trained for 100 epochs and the batch size is set to 200 for all the models.

The batch size during training and the dimension of the hidden states are two important hyperparameters of the proposed framework. The size of the training batch can make an influence on the final captioning results. The evaluation is carried out by calculating one parameter when other parameters are fixed. The parameter determination evaluation is conducted on UCM-captions data set. Experimental results for different batch sizes and hidden state dimension are shown in Figs. 4 and 5. To determine the proper size of training batch, the batch sizes are set to be 12, 25, 50, 100, and 200. As shown in Fig. 4, the best metrics are obtained when the batch size is 200. If the batch size continues to increase, the memory requirements of the computer will increase. Thus,

[2]https://pypi.org/project/python_speech_features/

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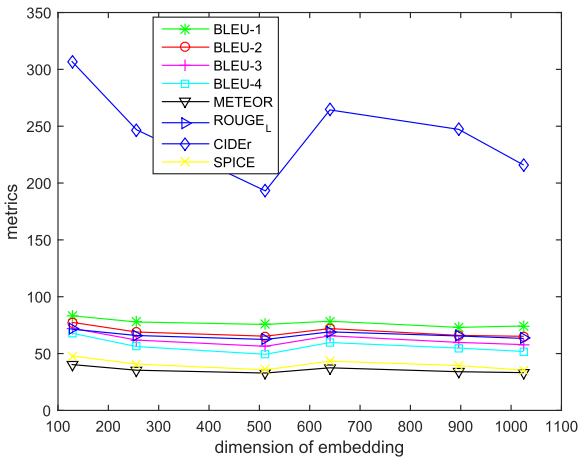


Fig. 5. Captioning metrics versus different dimensions of hidden states.

TABLE I
NUMBER OF MODEL'S PARAMETERS OF DIFFERENT DIMENSIONS OF HIDDEN STATES

dimension of hidden states	number of model's parameters
128	1029668
256	2556452
512	7379492
640	10675748
896	19037732
1024	24103460

the batch size is set to be 200 in the following experiments. The dimension of hidden states of the GRU can make an influence on the training time and captioning results. To determine the proper dimension of the hidden states, the dimensions of the hidden states are set to be 128, 256, 512, 640, 896, and 1024. As shown in Fig. 5, the best metrics are obtained when the dimension of the hidden states is 128. An increase in dimensions of hidden states can lead to an increase in the number of model's parameters. The relationship between the dimension of the hidden states and the number of the model's parameter is shown in Table I. To get the best results with the fewest parameters, the dimension of the hidden states is set to be 128, where the number of model's parameters is 1 029 668.

To make the CNN model more adapt to the data set, the model is fine-tuned on captioning data sets. The last two fully connected layers VGG-16 are removed. A fully connected layer with 1024 units is added to fine-tune the model. The learning rate of the fine tuning is 0.0001. The batch size is 10 and the training epochs are 100.

D. Experimental Results

In this section, the experimental evaluations of different methods on three HRRSI captioning data sets are summarized and discussed.

First experiment is conducted on the UCM-captions data set. As shown in Table II, the "-" indicates that the corresponding metric result is not available for that model. The black bold results denote the best result among four variants of the proposed framework. The "sound-a-f-ft" denotes the results after fine-tuning the pretrained CNN model. The mGRU-embedword in Table II means the word embedding layer is not trained from the captioning text. Instead, the word is embedded into a vector using a pretrained word model called GloVe [52]. As shown in Fig. 6, the mGRU gets convergence at about 40 epochs, but mGRU-embedword gets convergence at about 20 epochs. For the loss value, the loss value for the mGRU is finally reduced to about 1 and the loss value for the mGRU-embedword reduced to about 0.3. A conclusion can be made that the pretrained word model can make a faster convergence and better results. Therefore, the pretrained word model is adopted in the proposed framework.

To verify the fine-tuning strategy mentioned earlier, the sound-a-f-ft results, shown in Table II, are the metrics of the model after fine-tuning, using the strategy mentioned earlier. It can be seen that the fine-tuning of the model gives a metric boost on all the metrics. These results are consistent with the conclusion in [59]. Among the setting mentioned earlier, the sound-guided information is different for different images in the same class. A unique sound-guided information is given for one class. This means that the concentrate of the observer is the same part for one class. The unique sound-guided information for every class in UCM-captions data set is shown in Table III. Because it is hard for human to judge the corresponding words from the waveform of sound, the word of the sound is shown in text format. It can be observed from Table III that the word of sound is related to the class label following human common sense. For some classes, the word of sound is just corresponding to the class label, such as airplane, freeway, river, and runway. For some classes, the word of sound is directly related to the class label. Several examples are given in the following. The beach is consisted of sand. There are grasses in a piece of baseball diamond. The residential areas mainly consist of buildings and the "roofs" are the part of buildings that can be observed in a remote sensing image. The one-sound-one-class results, shown in Table II, are the metrics of the setting that a unique sound-guided information is given to all the images from one class. Table II shows that one-sound-one-class outperforms other methods according to all metrics except for METEOR and ROUGE_L.

In order to further verify the proposed framework, experiments are conducted on real-sound data sets. The experimental results are shown in the bottom part of Table II. It can be observed that the results are a little lower than that of fake sound because of the noise effect of real sound, but are still better than most other methods according to METEOR and BLEU-4.

The experimental performances of different methods on Sydney-captions data set are shown in Table IV. From the results, it can be observed that for the metric of BLEU, METEOR, and ROUGE_L, the proposed "f-a" framework gets the best scores among the method without attention mechanism. This is because the sound provides a better attention for the image. Proposed frameworks obtain higher performance than VLAD+RNN, VLAD+LSTM, and mRNN. The results of the proposed frameworks are better than mLSTM in terms of most metrics. The results of sound-f-a are comparable to

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TABLE II

CAPTIONING RESULT OF DIFFERENT METHODS ON UCM-CAPTIONS. THE SYMBOL "-" INDICATES THAT THE CORRESPONDING METRIC RESULT IS NOT AVAILABLE FOR THAT METHOD. THE BLACK BOLD RESULTS DENOTE THE BEST RESULTS AMONG FOUR VARIANTS OF THE PROPOSED FRAMEWORK. THE "SOUND-A-F-FT" DENOTES THE RESULTS AFTER FINE TUNING THE PRETRAINED CNN MODEL. THE "ONE-SOUND-ONE-CLASS" DENOTES THE RESULTS OF THE SETTING THAT ONE SOUND IS UTILIZED TO GUIDE ALL THE IMAGES IN ONE CLASS

UCM-captions with fake sound								
	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
VLAD+RNN [3]	63.11	51.93	46.06	42.09	29.71	58.78	200.66	-
VLAD+LSTM [3]	70.16	60.85	54.96	50.30	34.64	65.20	231.31	-
mRNN [1]	60.10	50.70	32.80	20.80	19.30	-	214.00	-
mLSTM [1]	63.50	53.20	37.50	21.30	20.30	-	222.50	-
mGRU [51]	42.56	29.99	22.91	17.98	19.41	37.97	124.82	30.58
mGRU-embedword	75.74	69.83	64.51	59.98	36.85	66.74	279.24	41.47
convcap [41]	70.34	56.47	46.24	38.57	28.31	59.62	190.15	29.48
soft-attention [3]	74.54	65.45	58.55	52.50	38.86	72.37	261.24	-
hard-attention [3]	81.57	73.12	67.02	61.82	42.63	76.98	299.47	-
sound-woattn	75.11	69.06	63.90	59.83	36.44	66.42	273.74	39.25
sound-a-f	**78.28**	**72.76**	**67.59**	**63.33**	**38.03**	**68.64**	**290.57**	**41.94**
sound-a-a	74.84	68.37	63.10	58.96	36.23	65.79	272.81	39.07
sound-f-f	72.07	65.32	59.37	54.71	34.15	62.96	249.62	37.61
sound-f-a	74.10	67.39	61.70	57.09	34.91	64.37	262.21	38.75
sound-a-f-ft	79.62	74.01	69.09	64.77	38.59	69.42	294.51	43.51
one-sound-one-class	83.26	77.48	72.28	67.92	40.33	71.47	306.87	47.87
UCM-captions with real sound								
sound-f-f-real	69.01	60.48	53.06	47.35	31.12	59.86	189.19	28.93
sound-f-a-real	74.12	67.65	62.03	57.42	35.07	65.42	262.18	39.53
sound-a-f-real	70.39	62.76	56.02	50.32	31.78	61.53	198.51	32.86
sound-a-a-real	69.95	59.23	50.53	43.74	30.73	59.82	180.93	29.95

TABLE III

ONE SOUND FOR ONE CLASS LIST FOR UCM-CAPTIONS

class	word of sound	class	word of sound	class	word of sound
agricultural	**cropland**	forest	**lots**	overpass	**roads**
airplane	**airplane**	freeway	**freeways**	parking lot	**lots**
baseball diamond	**grass**	golf course	**trees**	river	**river**
beach	**sand**	harbor	**boats**	runway	**runway**
building	**roofs**	intersection	**intersection**	sparse residential	**roofs**
chaparral	**lots**	medium residential	**houses**	storage tank	**storage**
dense residential	**roofs**	mobile home park	**homes**	tennis court	**court**

that of passive attention methods convcap according to SPICE. For other metrics, the results of passive attention methods are better than that of the proposed active attention framework. This may be because two kinds of attention mechanism have different emphases. The active attention mechanism is consistent with the observer's expectation, while the passive attention mechanism is to be close to the ground truth sentences.

To verify the fine-tuning strategy mentioned earlier, the sound-f-a-ft results, shown in Table IV, are the metrics of the model after fine tuning using the strategy mentioned earlier. Interestingly, it can be seen that the fine-tuning of the model gives a metric declination on all the metrics. Combining the phenomenon of fine-tuning on Sydney-captions and UCM-captions, it can be seen that the fine-tuning strategy is not always necessary, especially when the data set is small and unbalanced. Among the setting mentioned earlier, the sound-guided information is different for different images in the same class. A unique sound-guided information is given for one class. This means that the concentrate of the observer is the same part for one class. The unique sound-guided information for every class in Sydney-captions data set is shown in Table V. Because it is hard for human to judge the corresponding words from the waveform of sound, the word of the sound is shown in text format. It can be observed from Table V that the word of sound is related to the class label following human common sense. The one-sound-one-class results, shown in Table IV, are the metrics of the setting that a unique sound-guided information is given to all images from one class.

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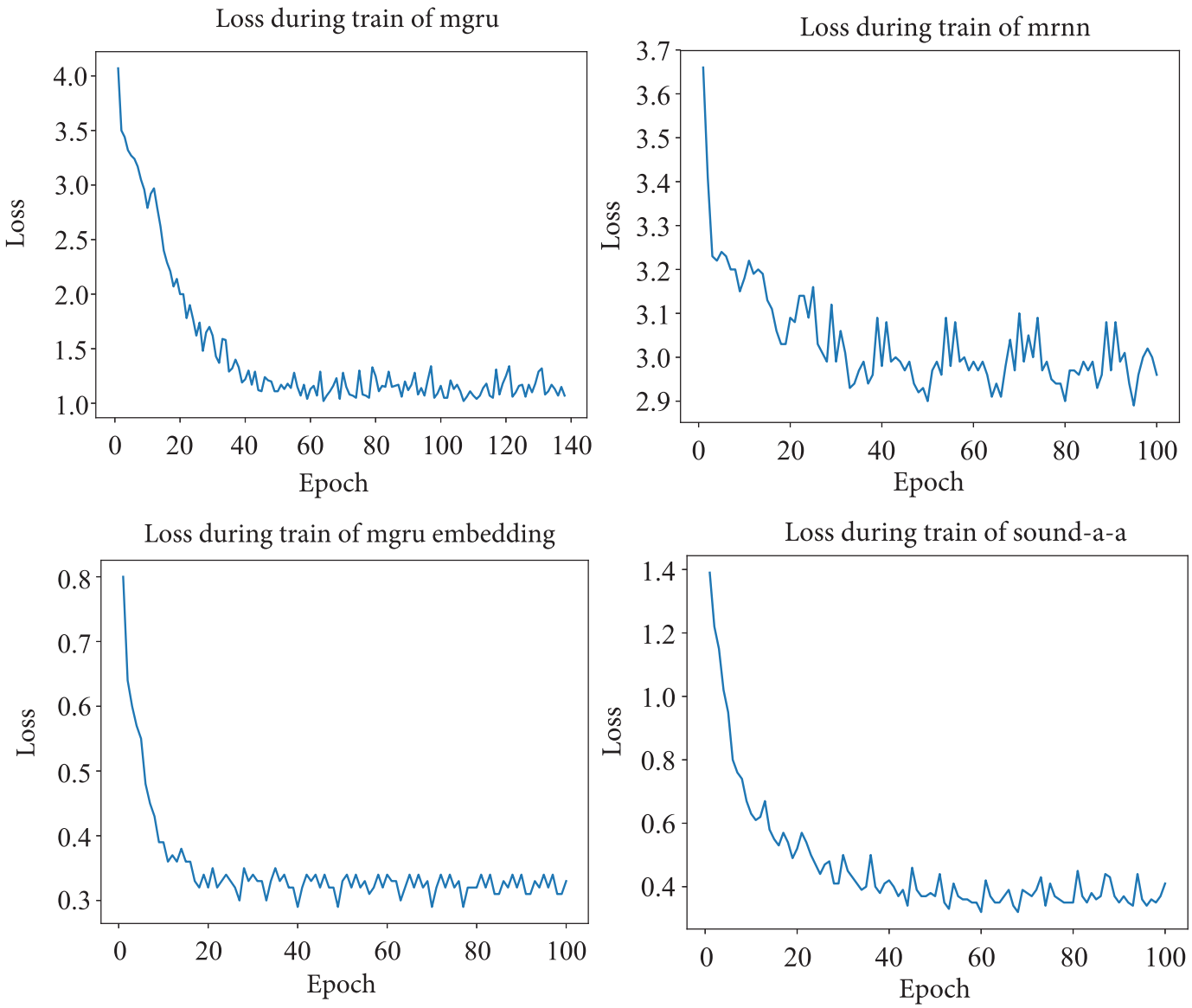


Fig. 6. Loss during training of different methods.

The experimental performances of different methods on RSICD data set are summarized in Table VI. It can be concluded that the proposed "a-a" framework obtains the best scores for the metric of BLEU and METEOR among almost all comparison methods except for passive attention methods. This may be because two kinds of attention mechanism have different emphases. The active attention mechanism is to accord with the observer's expectation, while the passive attention mechanism is to be close to the ground truth sentences.

To verify the fine-tuning strategy mentioned earlier, the sound-a-a-ft results, shown in Table VI, are the metrics of the model after fine-tuning using the strategy mentioned earlier. It can be seen that the fine-tuning of the model gives a metric boost on all the metrics. Among the setting mentioned before, the sound-guided information is different for different images in the same class. A unique sound-guided information is given for one class. This means that the concentrate of the observer is the same part for one class. The unique sound-guided information for every class in RSICD data set is shown in Table VII. Because it is hard for human to judge the corresponding words from the waveform of sound, the word of the sound is shown in text format. It can be observed from Table VII that the word of sound is related to the class label following human common sense. The one-sound-one-class results in Table VI are the metrics of the setting that a unique sound-guided information is given to all images from one class.

For fine-tuning strategy, the "ft" in Tables II–VI denotes the results of best framework after fine-tuning. It can be observed that the fine-tuning strategy gets a performance boost in UCM-captions and RSICD. But for Sydney-captions, the performance is declined. This is because the scale of the image number in Sydney is only 613, which is not enough to train a fully connected layer with 1024 units.

To explore the other format of the guiding information, the experimental results using text as the guiding information

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TABLE IV

CAPTIONING RESULT OF DIFFERENT METHODS ON SYDNEY-CAPTIONS. THE SYMBOL "-" INDICATES THAT THE CORRESPONDING METRIC RESULT IS NOT AVAILABLE FOR THAT MODEL. THE BLACK BOLD RESULTS DENOTE THE BEST RESULTS AMONG FOUR VARIANTS OF THE PROPOSED FRAMEWORK. THE "SOUND-F-A-FT" DENOTES THE RESULTS AFTER FINE TUNING THE PRETRAINED CNN MODEL. THE "ONE-SOUND-ONE-CLASS" DENOTES THE RESULTS OF THE SETTING THAT ONE SOUND IS UTILIZED TO GUIDE ALL THE IMAGES IN ONE CLASS

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
VLAD+RNN [3]	56.58	45.14	38.07	32.79	26.72	52.71	93.72	-
VLAD+LSTM [3]	49.13	34.12	27.60	23.14	19.30	42.01	91.64	-
mRNN [1]	51.30	37.50	20.40	19.30	18.50	-	161.00	-
mLSTM [1]	54.60	39.50	22.30	21.20	20.50	-	186.00	-
mGRU [51]	69.64	60.92	52.39	44.21	31.12	59.17	171.55	39.85
mGRU-embedword	68.85	60.03	51.81	44.29	30.36	57.47	168.94	37.01
convcap [41]	74.72	65.12	57.25	50.12	34.76	66.74	214.84	39.17
soft-attention [3]	73.22	66.74	62.23	58.20	39.42	71.27	249.93	-
hard-attention [3]	75.91	66.10	58.89	52.58	38.98	71.89	218.19	-
sound-woattn	69.11	60.73	52.37	44.42	30.69	58.73	172.09	39.07
sound-a-f	59.82	51.75	43.84	36.73	25.29	51.46	134.18	30.15
sound-a-a	70.93	62.28	53.93	46.02	31.21	59.74	174.77	38.37
sound-f-f	57.07	49.22	40.77	33.49	25.03	48.82	119.73	28.75
sound-f-a	**71.55**	**63.23**	**54.69**	**46.60**	**31.32**	**60.35**	**180.27**	**38.65**
sound-f-a-ft	68.82	60.73	52.94	45.39	30.49	58.20	170.52	37.86
one-sound-one-class	71.28	62.51	54.16	46.06	31.81	60.24	183.14	39.02

TABLE V

ONE SOUND FOR ONE CLASS LIST FOR SYDNEY-CAPTIONS

class	word of sound	class	word of sound	class	word of sound
residential	**houses**	airport	**airport**	meadow diamond	**lawn**
rivers	**river**	ocean	**ocean**	industrial	**containers**
runway	**runways**				

TABLE VI

CAPTIONING RESULT OF DIFFERENT METHODS ON RSICD. THE SYMBOL "-" INDICATES THAT THE CORRESPONDING METRIC RESULT IS NOT AVAILABLE FOR THAT MODEL. THE BLACK BOLD RESULTS DENOTE THE BEST RESULTS AMONG FOUR VARIANTS OF THE PROPOSED FRAMEWORK. THE "SOUND-A-A-FT" DENOTES THE RESULTS AFTER FINE TUNING THE PRETRAINED CNN MODEL. THE "ONE-SOUND-ONE-CLASS" DENOTES THE RESULTS OF THE SETTING THAT ONE SOUND IS UTILIZED TO GUIDE ALL THE IMAGES IN ONE CLASS

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
VLAD+RNN [3]	49.38	30.91	22.09	16.77	19.96	42.42	103.92	-
VLAD+LSTM [3]	50.04	31.95	23.19	17.78	20.46	43.34	118.01	-
mRNN [1]	45.58	28.25	18.09	12.13	15.69	31.26	19.15	10.94
mLSTM [1]	50.57	32.42	23.19	17.46	17.84	35.02	31.61	16.80
mGRU [51]	42.56	29.99	22.91	17.98	19.41	37.97	124.82	30.58
mGRU-embedword	60.94	46.24	36.80	29.81	26.14	48.20	159.54	34.77
convcap [41]	63.36	51.03	41.74	34.52	33.25	57.70	166.48	39.33
soft-attention [3]	67.53	53.08	43.33	36.17	32.55	61.09	196.43	-
hard-attention [3]	66.69	51.82	41.64	34.07	32.01	60.84	179.25	-
sound-woattn	56.98	41.69	32.17	25.41	23.65	45.08	108.92	29.01
sound-a-f	58.47	43.80	34.52	27.85	24.45	46.16	137.17	30.15
sound-a-a	**61.96**	**48.19**	**39.02**	**31.95**	**27.33**	**51.43**	**163.86**	**35.98**
sound-f-f	54.09	38.56	28.72	21.70	22.76	44.15	85.10	26.57
sound-f-a	59.35	45.11	35.29	28.08	26.11	49.57	132.35	32.65
sound-a-a-ft	67.60	54.30	44.33	36.45	31.09	55.36	193.96	41.74
one-sound-one-class	60.20	45.70	35.60	28.14	26.64	49.70	124.54	33.02

are shown in Tables VIII–X for UCM-captions, Sydney-captions, and RSICD, respectively. As shown in Table VIII, the results using sound as the guiding information are better than that using text as the guiding information on UCM-captions. In terms of text and sound, respectively, the result trend of different frame methods is the same. The frame methods are arranged according to the result from height to end as follows: a-f, a-a, f-a, and f-f. As shown

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TABLE VII
ONE SOUND FOR ONE CLASS LIST FOR RSICD

class	word of sound	class	word of sound	class	word of sound
airport	**airport**	farmland	**farmland**	playground	**playground**
bare land	**bareland**	forest	**forest**	pond	**pond**
baseball field	**baseballfield**	industrial	**industry**	port	**port**
beach	**beach**	meadow	**meadow**	railway station	**railway**
bridge	**bridges**	medium residential	**trees**	resort	**resort**
center	**buildings**	mountain	**mountain**	river	**river**
church	**church**	park	**park**	sparse residential	**pool**
commercial	**building**	school	**school**	storage tanks	**tanker**
dense residential	**building**	square	**square**	stadium	**stadium**
desert	**desert**	parking	**parking**	viaduct	**viaduct**

TABLE VIII
CAPTIONING RESULTS USING TEXT AND SOUND AS THE GUIDING INFORMATION ON UCM-CAPTIONS

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
text-a-f	73.59	66.99	61.66	57.51	35.10	64.51	264.23	38.70
sound-a-f	**78.28**	**72.76**	**67.59**	**63.33**	**38.03**	**68.64**	**290.57**	**41.94**
text-a-a	71.11	62.50	55.37	49.65	33.04	62.74	229.03	35.75
sound-a-a	**74.84**	**68.37**	**63.10**	**58.96**	**36.23**	**65.79**	**272.81**	**39.07**
text-f-f	64.90	56.61	49.96	44.71	27.91	57.65	200.53	32.52
sound-f-f	**72.07**	**65.32**	**59.37**	**54.71**	**34.15**	**62.96**	**249.62**	**37.61**
text-f-a	72.02	63.28	54.88	48.26	32.27	62.96	192.23	33.96
sound-f-a	**74.10**	**67.39**	**61.70**	**57.09**	**34.91**	**64.37**	**262.21**	**38.75**

TABLE IX
CAPTIONING RESULTS USING TEXT AND SOUND AS THE GUIDING INFORMATION ON SYDNEY-CAPTIONS

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
text-a-f	**69.84**	**61.01**	**52.57**	**44.47**	**30.47**	**59.17**	**171.09**	**37.22**
sound-a-f	59.82	51.75	43.84	36.73	25.29	51.46	134.18	30.15
text-a-a	**71.31**	**62.31**	**54.09**	**46.26**	**31.69**	**60.21**	**179.20**	**38.52**
sound-a-a	70.93	62.28	53.93	46.02	31.21	59.74	174.77	38.37
text-f-f	**66.06**	**56.97**	**48.25**	**40.36**	**27.99**	**55.62**	**147.57**	**34.80**
sound-f-f	57.07	49.22	40.77	33.49	25.03	48.82	119.73	28.75
text-f-a	70.88	62.26	53.76	45.56	31.25	59.87	175.54	**39.01**
sound-f-a	**71.55**	**63.23**	**54.69**	**46.60**	**31.32**	**60.35**	**180.27**	38.65

TABLE X
CAPTIONING RESULTS USING TEXT AND SOUND AS THE GUIDING INFORMATION ON RSICD

	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr	SPICE
text-a-f	53.62	37.92	27.72	20.56	21.78	42.37	79.51	24.85
sound-a-f	**58.47**	**43.80**	**34.52**	**27.85**	**24.45**	**46.16**	**137.17**	**30.15**
text-a-a	**65.05**	**51.32**	**41.44**	**33.61**	**29.19**	**52.68**	**168.28**	**38.85**
sound-a-a	61.96	48.19	39.02	31.95	27.33	51.43	163.86	35.98
text-f-f	52.86	37.78	28.59	22.05	22.69	44.02	90.14	25.38
sound-f-f	**54.09**	**38.56**	**28.72**	**21.70**	**22.76**	**44.15**	**85.10**	**26.57**
text-f-a	**61.30**	**47.54**	**37.74**	**30.22**	**27.31**	**50.02**	**140.65**	**34.96**
sound-f-a	59.35	45.11	35.29	28.08	26.11	49.57	132.35	32.65

in Table IX, the results using sound as the guiding information are better than that using text as the guiding information on Sydney-captions only for the frame f-a according to all the metrics except for SPICE. As shown in Table X, the results using sound as the guiding information are better than that using text as the guiding information on RSICD only for the

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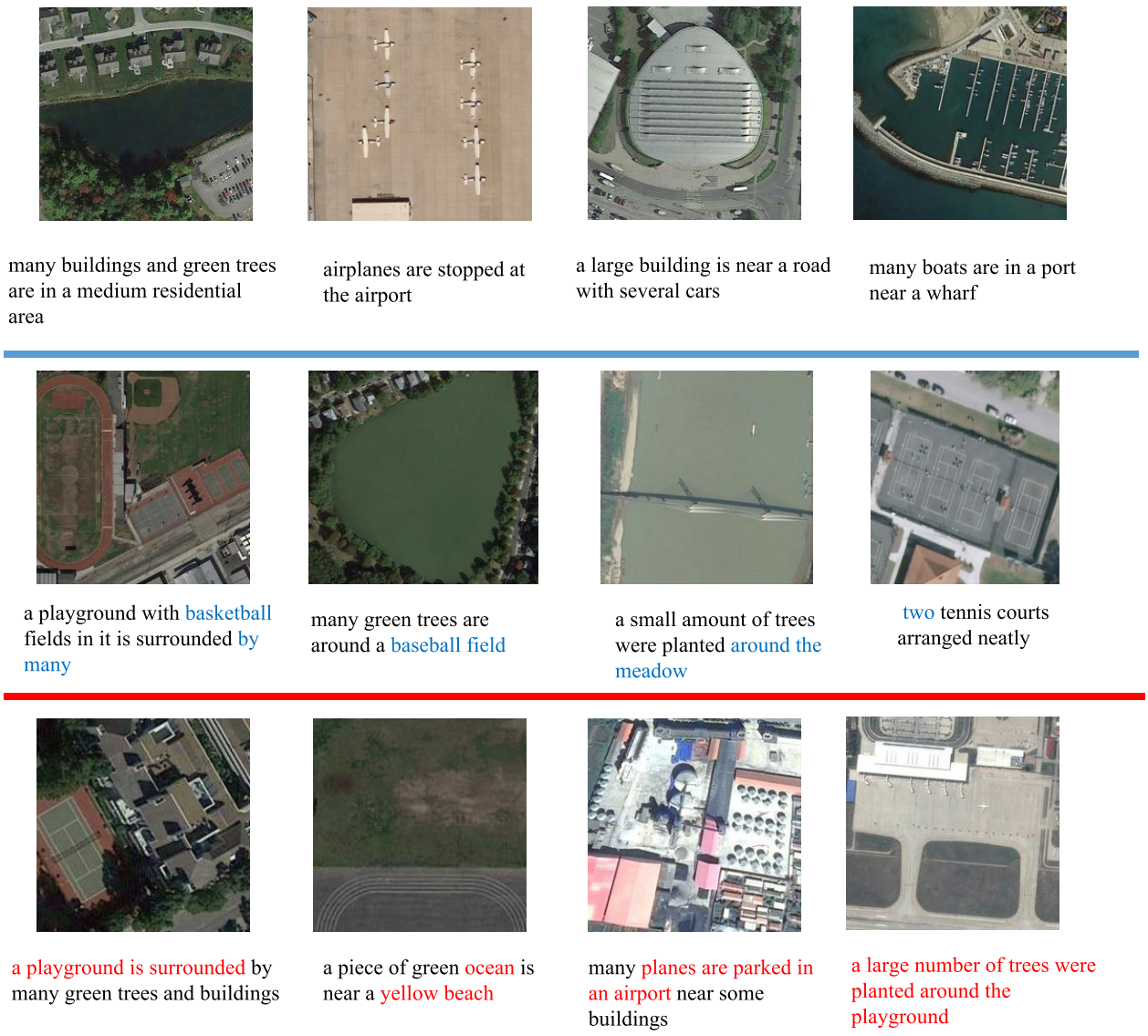


Fig. 7. Captioning examples of different levels of human evaluation. First row: examples that are totally right. Second row: examples that are partly right. Third row: examples that are totally wrong. The words in blue in the second row and the words in red in the third row are words that are not matched with the corresponding image.

frames f-f and a-f. Different from the experimental results on UCM-captions, the performance of the framework using text as the guiding information is better than that using sound in most cases for Sydney-captions and RSICD. This may be because the characteristics of data sets are different from data scale to content distribution. The data scale difference can be seen in Section IV-A1. As for content distribution, specifically, for text features, the feature length is the same when one hot is used for feature coding. But for sound, the longer the sound file is, the longer the MFCC feature is, and the more difficult the useful information extraction is. The one-hot representation of text guiding is robust and the results of text guiding can be seen as a reference of sound guiding. In order to count the length of audio files in different data sets, the length of sound features is reflected by counting the number of letters contained in words. For UCM-captions, the average length of letters for every image is 5.53, which is smaller than 5.79 of Sydney-captions and 5.71 of RSICD. These long audio files make the sound information encoding difficult for Sydney-captions and RSICD. Thus, the performance of the framework using text as the guiding information is better than using sound for Sydney-captions and RSICD. To sum up, the frame methods using sound as the guiding information have some advantages in some cases. As the sound and text contain the same semantic content, there is no absolute difference in performance. Because of the convenience and rapidity of sound input, the sound is adopted as the guiding information.

The objective metric scores mentioned above can reflect the correlation between the image and the sentence implicitly. A better way to evaluate is to let volunteers look the captioning results and give a human evaluation score. The level of human evaluation score is divided into three levels according to the correlation between the image and the sentence: totally right, partly right, and totally wrong. The statistical results of three data sets are shown in Fig. 8. It can be observed that the

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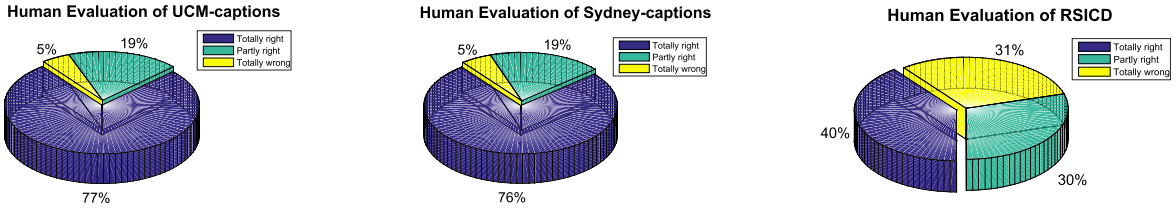


Fig. 8. Human evaluation of different data sets.



trees: a playground is surrounded by many green trees and buildings.
industry: a large number of cars parked in the airport
ponds: a large number of tall buildings have been built
factory: a large number of trees were planted around the factory



railway: many buildings and green trees are in a school
station: many buildings are in a commercial area
rails: many buildings and green trees are in a school
grey: many buildings and green trees are in a medium residential area



pool: many green trees and a building are in a resort near a road
triangle: many green trees are around a baseball field
road: many green trees and a road are around a park with a pond
trapezoidal: many green trees are around a building with a swimming pool



airport: many planes are parked near a terminal in an airport
highway: many planes are parked near several terminals in an airport
aeroplane: many planes are parked near several terminals in an airport
church: many buildings are around a church



color: many buildings and green trees are in a school with a pond
pond: many buildings are around a square with many green plants
trigngle: many planes are parked in an airport near several buildings
trapezoidal: many buildings and green trees are in a medium residential area



factory: many planes are parked in an airport near some buildings
storage: many buildings are around a park with a pond
land: many buildings are around a park with a pond
road: many buildings and green trees are near a road

Fig. 9. Captioning examples of different captioning results based on different input sounds. The word in bold before the sentence is the text of the input sound.

almost 80% of the generated sentences are totally right for UCM-captions and Sydney-captions. For RSICD, 40% of the generated sentences are totally right.

To give readers a more intuitive results, some captioning examples are shown in Fig. 7. Three levels of human evaluation are listed in different rows: totally right example in the first row, partly right example in the second row, and totally wrong example in the third row. The volunteers to judge the captioning results may not participate in the construction of the data set. Sometimes even the sentence is not corresponding to the "right" sentence in the data set, the volunteers give a "totally right" score. As shown in Fig. 7, the main topic of the first image should be the green pond in triangular. But next to the pond, there is a parking lot and a medium residential as well. Most evaluation volunteers think the sentence is "totally right." For some wrong places, it can be observed that the second image in the second row is considered to be a baseball field. But actually, it is a pond in almost triangular shape, which is the same shape of a baseball field. The number of tennis courts is wrong for the fourth image in the second row. For the third image in the third row, the factory with many small storage tanks looks likes an airport with many planes, which is totally wrong. It can be observed that most wrong examples are related to the appearance representation of image. This is related to the scale of the data set. Neural networks need a large amount of data to train a representation that can distinguish the similar images. The sound-guided captioning results get a better performance than the original captioning task overall.

In order to show that different sound-guided information can generate different sentences that concentrate on different concepts, some examples are listed in Fig. 9. The black bold words before the sentence is the word of the sound. As shown in Fig. 9, the different sound-guided information can generate different sentences. Take the first image in the first row in Fig. 9 as an example, and the content of the image is a factory with buildings and trees. When the focus is the trees, the generated sentence is more related to trees. But when the focus is the factory, the sentence can capture the factory. It should be observed that there are some wrong places. For example, when the focus is industry, the sentence is wrong. This is because the buildings in a factory are similar with

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TABLE XI
TRAINING AND TESTING TIME OF DIFFERENT METHODS ON UCM-CAPTIONS. M MEANS MINUTE AND S MEANS SECOND

	Training time (m)	Testing time (s)
VLAD+RNN [3]	121.43	2.57
VLAD+LSTM [3]	291.18	2.89
mRNN [1]	18.64	6.39
mLSTM [1]	23.58	5.83
mGRU [51]	34.00	32.43
mGRU-embedword	31.75	29.76
convcap [41]	1567.32	56.21
soft-attention [3]	1251.51	140.02
hard-attention [3]	1310.45	140.79
sound-woattn	30.10	27.94
sound-a-f	38.08	37.01
sound-a-a	38.43	36.29
sound-f-f	37.80	36.07
sound-f-a	37.89	38.30

buildings in an airport. Another interesting phenomenon is that the buildings can be correctly depicted even when the sound is a wrong disturb information "ponds." More examples can be seen in Fig. 9.

E. Training and Testing Time

The experiments are conducted on Ubuntu 14.04.5 LTS with 48 Intel(R) Xeon(R) CPU E5-2650 v4 at 2.20 GHz. Training and testing times are important indicators to measure the efficiency of different methods. The training and testing times of different methods on UCM-captions are reported in Table XI. The convcap [41] directly optimizes the evaluation CIDEr, while other algorithms optimize the cross-entropy loss function. It takes longer to calculate CIDEr than to calculate cross entropy. mRNN [1], mLSTM [1], mGRU [51], mGRU-embedword, and the proposed method are implemented with Keras, which is a relatively fair comparative premise. The training time of the sound-woattn is close to that of mGRU-embedword. This is because the basic structure of both algorithms is GRU. After adding attention, the training time rises from 30 to about 38 min. This phenomenon is caused by the increase of parameters introduced by attention. For testing time, the testing time of sound-a-f is increased from 27.94 to 37.01 s compared with that of sound-woattn due to the introduction of attention.

V. CONCLUSION

In this article, sound is introduced into the captioning task to capture the specific objects that an observer is interested in. Different from the traditional captioning task, sound-guided captioning is to generate the descriptive sentence given the original image and the sound that a person given representing the person's concept of interest. A novel sound active attention framework is proposed to combine the sound during the generation of sentence given an image. There are four different variants of the framework and these variants are evaluated on three HRRSI captioning data sets. The contributed experiments show that the proposed methods can generate sentences that capture the concept of interest of an observer.

Furthermore, the valuable future research on using the active attention mechanism to guide the caption generation of remote sensing images is mainly divided into two aspects: one is the potential active attention in other tasks, such as automatic drive, and the other is potential formats of active attention. Specifically, the design of the attention method can also draw lessons from the attention mechanism in the ordinary attention method or design a more exquisite active attention structure.

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